Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech
Code: 100507

Subject: Power Systems-I (Apparatus and Modelling)

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Corona loss can be reduced by using
(i) Bundle conductor
(ii) Solid conductor
(iii) Hollow conductor
(i) (i) only (ii) (ii) and (iii) only
(iii) (i) and (iii) only (iv) All of the above

(b) The rated voltage of a 3 phase power system is given by
(i) Rms phase value (ii) Rms line to line voltage
(iii) Peak phase voltage (iv) Peak line to line voltage

(c) In which climate does the changes of occurrence of corona is maximum?
(i) Damp weather (ii) Hot summer
(iii) Winter season (iv) Cold and dry weather

(d) The most suitable location for the power factor improvement device is
(i) Anywhere in the power system it will provide same effect. (ii) Near the electrical appliance which is responsible for the poor power factor
(iii) Near the generating end (iv) At the sending end.

(e) On the basis of reliability which topology is more reliable
(i) Radial system (ii) Meshed system
(iii) Both have same reliability (iv) Realibility of power system will not change with topology

(f) Length of the cable is doubled, its capacitance C will be
(i) Doubled (ii) Halved
(iii) Will remain unchanged (iv) Will be 1/4th.

(g) A 3-phase 4 wire system is used for
(i) Primary transmission (ii) Secondary transmission
(iii) Distribution (iv) In all three above parts of power system

(h) Skin effect of conductor has following effect
(i) Reduced DC resistance (ii) Reduced AC resistance
(iii) Increased DC resistance (iv) Increased AC resistance

(i) For a long distance high voltage transmission line at under loaded condition the receiving end voltage is _____ the sending end voltage.
(i) Equal to (ii) Less than
(iii) Greater than (iv) Zero irrespective of

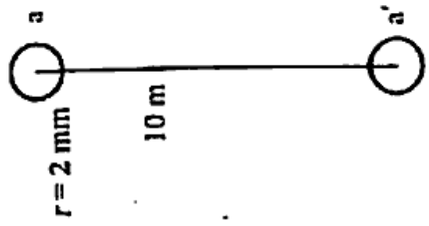
(j) The conductors that connect the distribution pole with the consumer is known as
(i) Distributor (ii) Feeder
(iii) Service main (iv) Transmission line

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Q.2 (a) For an isolated conductor determine the total inductance (for both internal and external flux linkage) of the conductor. [8]

(b) What is transposition of transmission line. Show the transposition of transmission line by schematic diagram, labelling all three phases for all three sections [6]

Q.3 (a) [7]



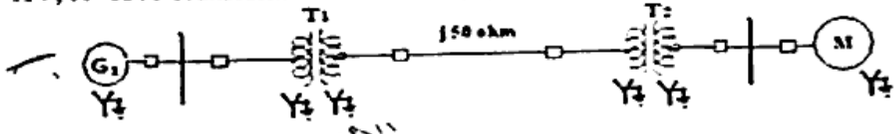
The figure Shows a two conductor single phase system. The conductors are separated by a distance of 10m and have same cross-sectional area with radius of 2mm. Determine inductance per km of the system.Calculate system's total inductive reactance when frequency is 50Hz and length of the line is 10km.

(b) Derive the expression of capacitance of a two-wire isolated line. [7]

Q.4 Draw the P-V and I-V characteristics of PV panels and explain. [14]

Q.5 (a) Write three most important advantages of representing power system quantities by PU system of representation. [6]

(b) Draw the per unit reactance diagram for the power system shown in figure below. The ratings of the generator motor and transformer are: Generator: 50MVA, 30KV, X=20% Motor: 50MVA, 11KV, X=30% Transformer T1: 40MVA, 33/220 KV, X=15% Transformer T2: 30MVA, 220/11 KV, X=15%. [8]



Q.6 (a) What is corona loss. Describe corona effect. [6]

(b) List the factors that increase the chance of corona occurance. [8]
What is the effect of smoothness of conductor surface on the event of corona?

Q.7 Draw any electrical equivalent model of medium transmission line (the nominal T or nominal π model representation) and derive the expressions for A, B, C, D parameters for the model. Clearly mention the unit of each of these four parameters. [14]

Q.8 (a) A 3-phase, 50-Hz 50 km long overhead transmission line has the resistance/km/phase = 0.1 ohm, inductive reactance/km/phase = 0·2 ohm. Determine [10]
(i) the receiving end current
(ii) sending end current
(iii) sending end voltage
(iv) transmission efficiency, when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0·8 lagging.

(b) A 50 Hz transmission line (200 km long) has a total series impedance of 40 + j125 ohms and total shunt admittance of 10-3 mho. The receiving-end load is 50 MW at 220 kV with 0.8 lagging power factor. Find the sending-end voltage and current using nominal-π representation method. [4]

Q.9 short notes on *any two* of the following: [7 x 2 = 14]
(a) Insulation coordination
(b) Voltage source converters
(c) Neutral grounding of transformer

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